



Sort kar do array



inc ↑ sort karna

Sorting Algorithms

arr = { 1, 0, -1, 2, 9, 3, 7 }



{ -1, 0, 1, 2, 3, 7, 9 }



Contents

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Ques: Check if array is sorted

$arr = \{ 1, 0, 2, 3, 4, 5, 6 \}$ false

$arr = \{ -1, 0, 1, 2 \}$ true

$arr = \{ 1, 1, 2, 3, 4, 4 \}$



Bubble Sort → ek algo hai jisse array sort hota hai

arr = { 3, 5, 1, 4, 2, 0 }

arr = { 0, 1, 2, 3, 4, 5 }

'n-1' passed honge

Bubble Sort (thoda better)



$arr = \{1, 2, 3, 4, 5\}$

before each pass, we can actually check if the array is sorted or not.



Bubble Sort (Optimised) ✓

Time Complexity

Best Case : $O(n)$

Avg. Case : $O(n^2)$

Worst Case : $O(n^2)$

Bubble Sort (Reverse)

Q1: Sort an array in descending order using bubble sort.



Homework.

Ques: Move all Zeros to end

$arr = \{ 1, 0, -2, 3, 0, 4, 8, 0, 10, 12 \}$



$arr = \{ 1, -2, 3, 4, 8, 10, 12, 0, 0, 0 \}$

Ques: Move all Zeros to end

arr = { 1, -2, 3, 4, 8, 10, 12, 0, 0, 0 }

j
 i

Built-in Sort (Merge Sort)

↓

T.C. = $O(n \log n)$

A.S. = $O(n)$



`Arrays.sort(arr)`

`Collections.sort(list)`



Selection Sort (Select Smallest)

arr = { 8, 4, 1, 4, -3, 6, 5 }

arr = { -3, 4, 1, 4, 8, 6, 5 }

arr = { -3, 1, 4, 4, 8, 6, 5 }

arr = { -3, 1, 4, 4, 8, 6, 5 }

arr = { -3, 1, 4, 5, 8, 6, 9 }

arr = { -3, 1, 4, 5, 6, 8, 9 }

arr = { -3, 1, 4, 5, 6, 8, 9 }

T.C.

Best: $O(n^2)$

Avg: $O(n^2)$

Worst: $O(n^2)$

T.n.o.

$= n + n-1 + n-2 \dots 1$

$= \frac{n(n+1)}{2}$

Selection Sort (find largest first)

Homework

arr = { 8, 4, 1, 9, -3, 6, 5 }

arr = { 8, 4, 1, 5, -3, 6, 9 }

⋮



Ques: 2 Sum - Find a pair with given sum

arr = { 7, 0, 4, 3, 2, 8, 10 } target = 9

↓
sort $\rightarrow O(n \log n)$

{ 0, 2, 3, 4, 7, 8, 10 } $\rightarrow O(n)$
 i *j*

$$O(n \log n + n) = O(n \log n)$$



Ques: 2 Sum - Find a pair with given sum

arr = { 7, 12, 26, 41, 54 } target = 50

j
i

Stability of Bubble & Selection Sort

arr =

7 ₁	3	4	7 ₂	8	1
----------------	---	---	----------------	---	---

↙
bubble Sort

↓
Selection Sort

1	3	4	7 ₁	7 ₂	8
---	---	---	----------------	----------------	---

Stable

1	3	4	7 ₂	7 ₁	8
---	---	---	----------------	----------------	---

unstable



Ques: Common Elements

$$a = \{3, 1, 2, 1, 1, 4, 5, 5\}$$

↓ Sort

$$b = \{6, 1, 1, 4, 4, 2, 8\}$$

↓ Sort

$$\rightarrow a = \{1, 1, 1, 2, 3, 4, 5, 5\} \quad b = \{1, 1, 2, 4, 4, 6, 8\}$$

ij

$$\text{ans} = \{1, 1, 2, 4\}$$



HW: Union of 2 sorted Arrays



HW: Intersection of Arrays with Distinct



Ques: Kth smallest element



Insertion Sort

more similar to B.S.

Adjacent elements swapping

arr = { 4 1 7 3 9 1 0 8 }

arr = { 0 1 1 3 4 7 8 | 9 }

i

j

there are 2 parts of array, sorted part & unsorted part.

Unsorted part ka pehla element sorted part me sahi jagah insert karna hai

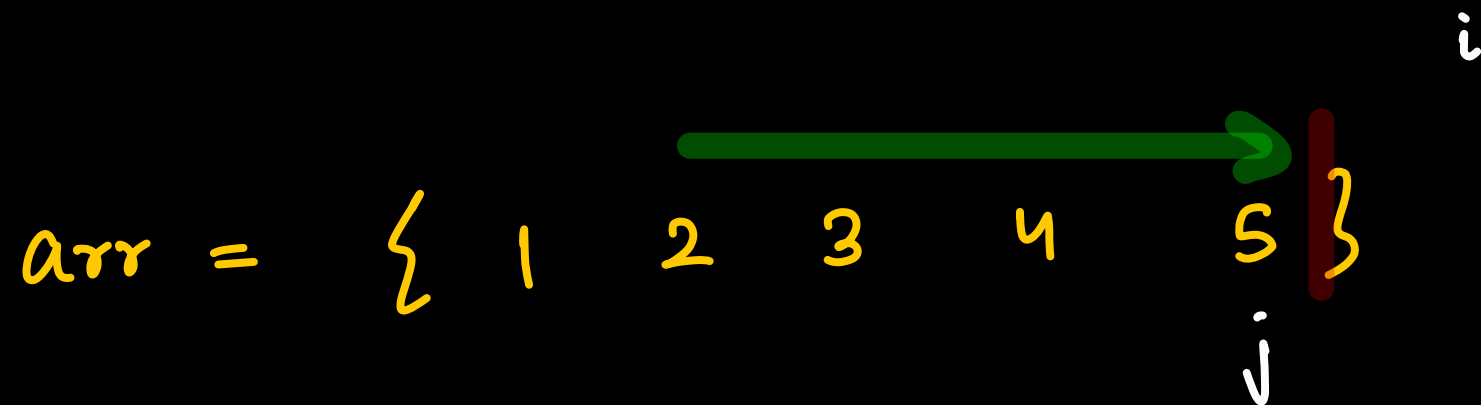
Time Complexity of Insertion Sort

* Best Case $O(n)$ Array is already sorted

Average Case $O(n^2)$

Worst Case $O(n^2)$ Array is sorted in desc order

Auxiliary Space $O(1)$



$$\text{Swaps} = 1 + 2 + 3 + 4 + \dots + n-1 = \frac{n(n-1)}{2}$$



Stability of Insertion Sort

arr = {4, 1, 7, 3, 9, 1, 0, 8}

0 1 1₂ 3 4 7 8 9

Insertion sort is a stable sort



THANKYOU

Cuties